Line impedances

There are two impedances defined for transmission lines,

1. Characteristic impedance &
2. input impedance

These impedances are defined and formulated as follows:

a. Lossy line:

If the transmission line has a very long length $l \to \infty$, the incident voltage component becomes $V_i e^{-\gamma z} \to 0$ at the load end and thus the reflection of the signal does not arise.

When the reflected wave in the line is zero, the ratio at any point z, $V(z)/I(z)$ is called the "characteristic-impedance" of the line and is defined by

$$Z_0 = \frac{V(z)}{I(z)} = \frac{V_i e^{-\gamma z}}{I_i e^{-\gamma z}} = \frac{V_i}{I_i} = \sqrt{\frac{Z}{Y}} = \sqrt{\frac{R+j\omega L}{G+j\omega C}}$$

$$= R_0 + jX_0 \ \Omega \quad \text{---} (1)$$

Where R_0 and X_0 are the real and imaginary parts of Z_0.

When $R \ll \omega L$ and $G \ll \omega C$ for low-loss lines and also for microwave frequencies,

$$Z_0 = \sqrt{L/C} \text{ and is real.} \quad \text{---} (2)$$

The reciprocal of Z_0 is called the characteristic admittance Y_0

i.e., $Y_0 = 1/Z_0$.

From fig below:

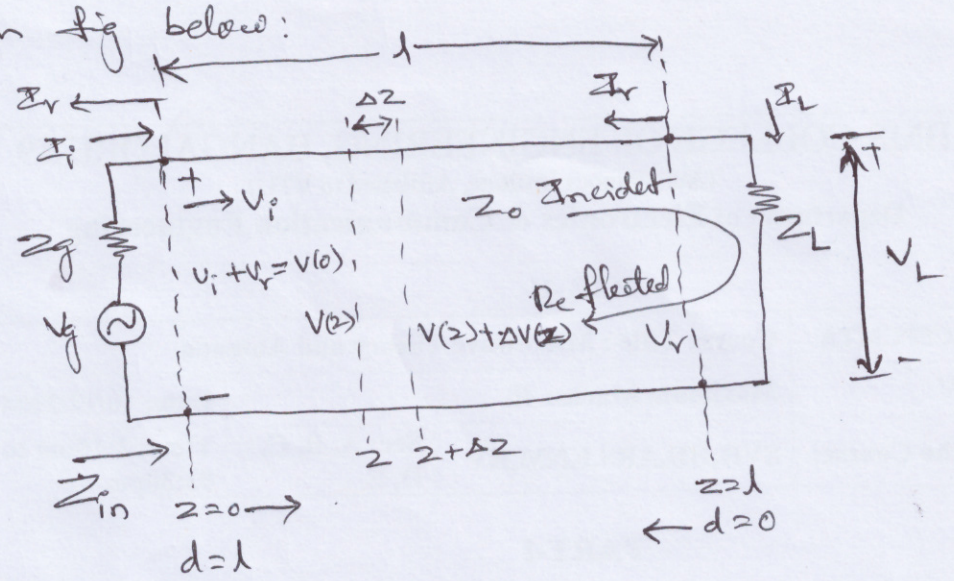


fig: General schematic of transmission line

It is seen that the impedance of the line at any point 'z' - looking towards the load ~~ends~~ end can be generally expressed by

$$Z_{in}(z) = \frac{V(z)}{I(z)} = \frac{V_i e^{-\gamma z} + V_r e^{\gamma z}}{I_i e^{-\gamma z} + I_r e^{\gamma z}}$$

$$= Z_0 \frac{V_i e^{-\gamma z} + V_r e^{\gamma z}}{V_i e^{-\gamma z} - V_r e^{\gamma z}} \quad \text{——(3)}$$

Here $I(z) = V(z)/Z_0$ ~~and~~ is replaced by $V(z)/Z_0$ in the above Equation.

⇒ The "-ve" sign in the denominator arises due to reflected current direction opposite to that of the incident current.

The above circuit can be replaced by the Equivalent circuit shown below.

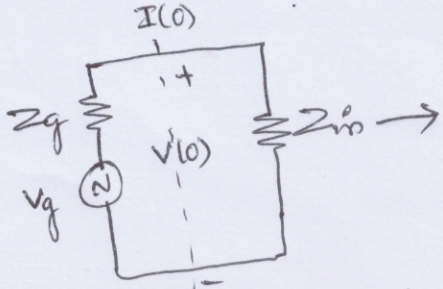


fig: Equivalent circuit of transmission line

The unknown values V_i, V_r can be determined from the terminal conditions at $z=0$ and $z=l$: $V(0)$ and $I(0)$.

Substituting these values to the basic eqns below

$$V(z) = V_i e^{-\gamma z} + V_r e^{\gamma z} \quad \text{— (A)}$$

$$I(z) = I_i e^{-\gamma z} + I_r e^{\gamma z} \quad \text{— (B)}$$

which gives

$$V_i = \tfrac{1}{2}[V(0) + Z_0 I(0)] \quad \text{— (4)}$$

$$V_r = \tfrac{1}{2}[V(0) - Z_0 I(0)] \quad \text{— (5)}$$

From the equivalent circuit

$$V(0) = \frac{V_g Z_{in}}{Z_{in} + Z_g} \quad \text{and} \quad I(0) = \frac{V_g}{Z_{in} + Z_g} \quad \text{— (6)}$$

Hey, at load end, $I(l) = I_L$ and $V(l) = V_L = I_L Z_L$;

substituting in basic eqns (A) & (B),

$$V_i = \tfrac{1}{2}(V_L + Z_0 I_L) e^{\gamma l} \quad \text{and}$$

$$V_r = \tfrac{1}{2}(V_L - Z_0 I_L) e^{-\gamma l} \quad \text{— (7)}$$

Here, ~~$V_L = I_L / Z_L$~~ . $V_L = I_L \cdot Z_L$

From eqn (3) to eqn (7), substituting the values of V_i and V_r and using the definition of sine and cosine hyperbolic - functions in terms of exponential function

i.e., $\sinh x = \frac{e^{-x} + e^{x}}{2}$ and $\cosh x = \frac{e^{-x} - e^{x}}{2}$,

the general expression of input impedance of a length 'l' of line at the generator end is given by

$$Z_{in}(0) = Z_0 \frac{Z_0 \sinh \gamma l + Z_L \cosh \gamma l}{Z_L \sinh \gamma l + Z_0 \cosh \gamma l} = Z_0 \frac{Z_L + Z_0 \tanh \gamma l}{Z_0 + Z_L \tanh \gamma l} \quad \text{— (8)}$$

(i) If the line is short circuited, $Z_L = 0$ & $V_L = 0$

$$\therefore Z_{in\,sc} = Z_0 \tanh \gamma l$$

(ii) For an open circuited line, $Z_L = \infty$, $I_L = 0$

$$Z_{in\,oc} = Z_0 \coth \gamma l$$

(b) Lossless lines:

For a lossless line, $\alpha = 0$, $\gamma = j\beta$

$\tanh \gamma l = \tanh(j\beta l) = j \tan \beta l$, then eqn (8) reduces to

$$Z_{in}(0) = Z_0 \frac{Z_L + jZ_0 \tan \beta l}{Z_0 + jZ_L \tan \beta l} \quad \text{---(9)}$$

$$\beta = 2\pi/\lambda.$$

The quantity 'βl' is called the electrical length in radians of the line.

(i) For short circuited line

$$Z_{in\,sc} = jZ_0 \tan \beta l$$

(ii) For open circuited line

$$Z_{in\,oc} = -jZ_0 \cot \beta l.$$